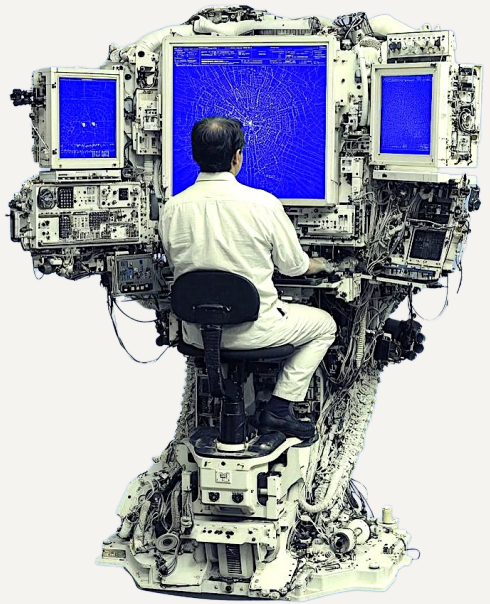
NUS-ISS · PATTERN RECOGNITION SYSTEMS

MACHINESENSE

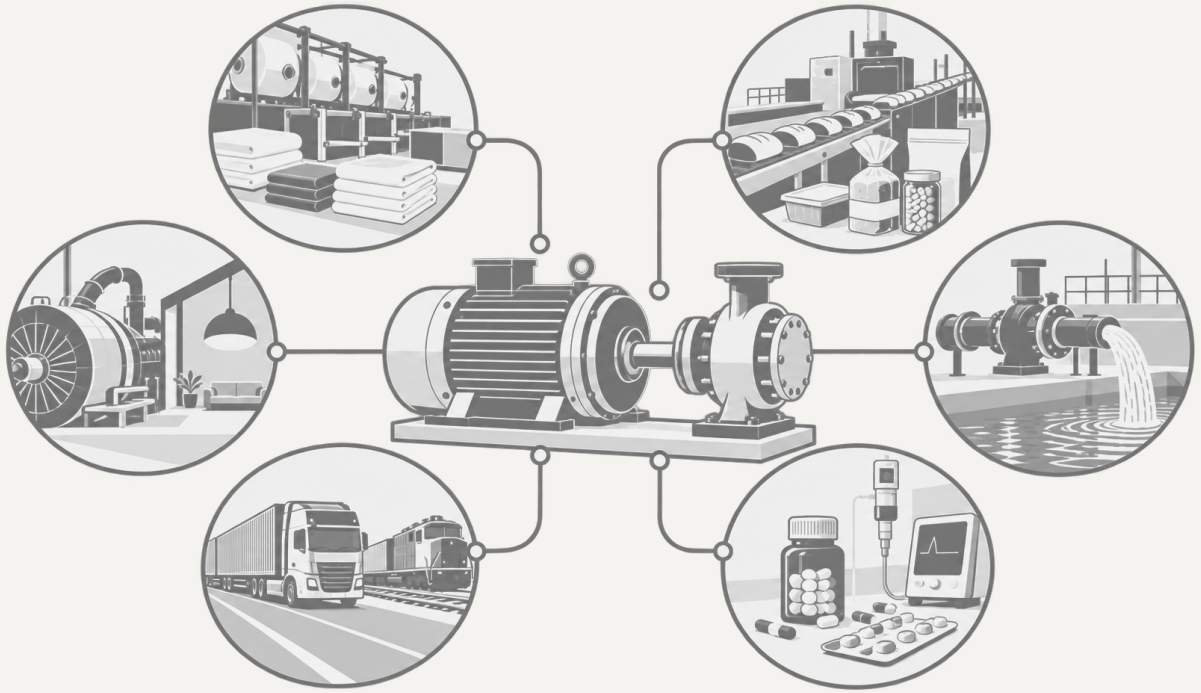
Multi-sensor fault diagnosis for **rotating machinery**, from a five-second reading to an evidence-supported inspection decision.

Bhuvesh Kumar · A0243823H
MTech AI Systems · Practice Module



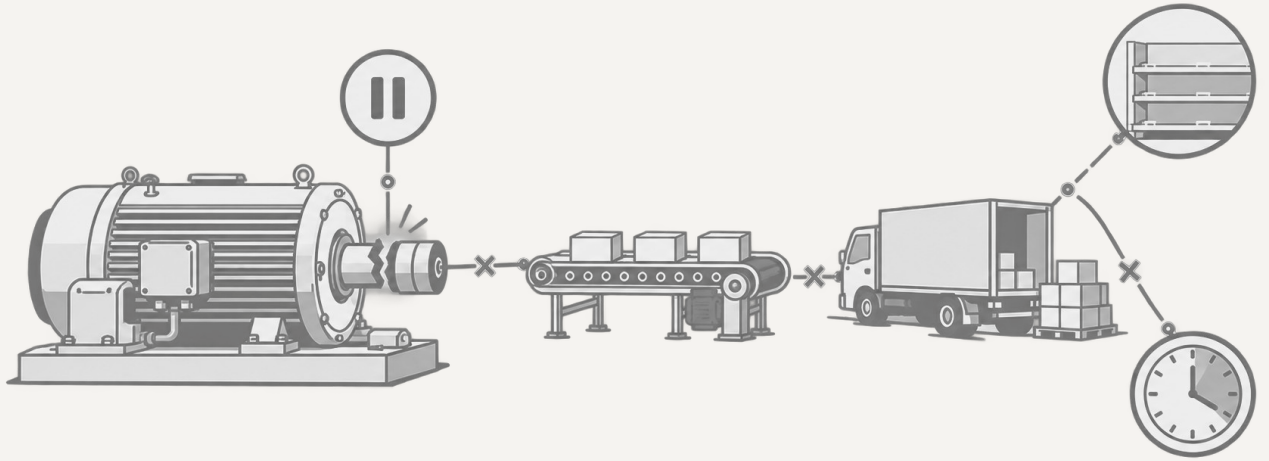
MOTIVATION

Machines keep our lives moving



The clothes we wear, the food on the shelf, the power in the socket, the water in the tap, everything that moves and every hospital that stays open all run on rotating machinery.

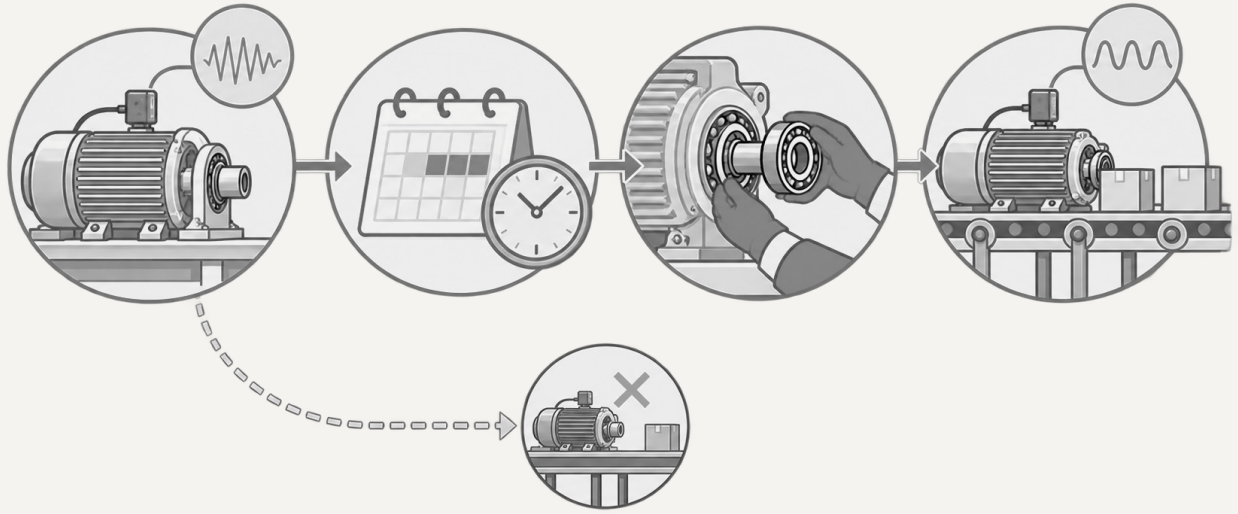
What happens when a machine breaks down without warning?



One bearing gives way and the line stops. Production halts, deliveries are late, shelves empty, and the repair costs far more than it would have before the failure.

PREVENTIVE MAINTENANCE

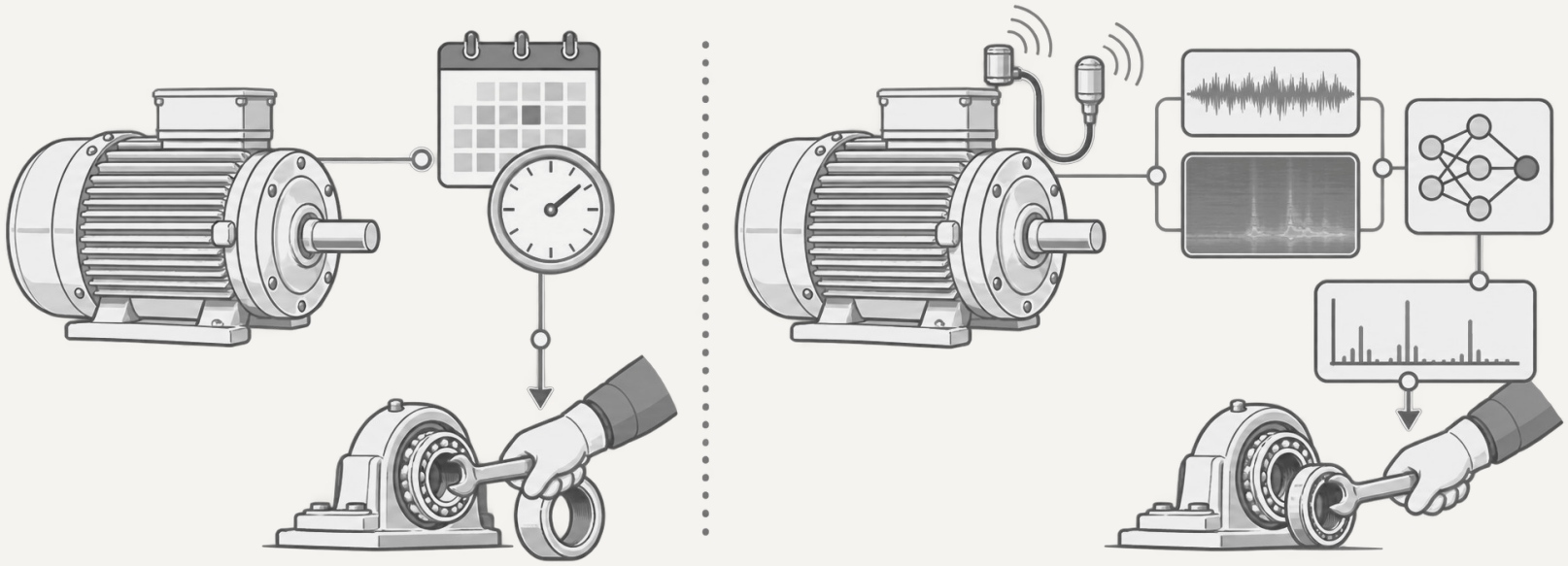
Most of this is preventable



Measure the machine while it is still running, plan the work, replace the part before it fails, and the machine goes back into service. Preventive maintenance turns a breakdown into a scheduled job.

CLASSIC VERSUS MACHINE LEARNING

Two ways to decide when to intervene



Classic preventive maintenance replaces parts on a fixed schedule, so healthy parts are discarded and some faults still arrive early. A machine-learning approach listens to the machine itself and acts on what the signal says.

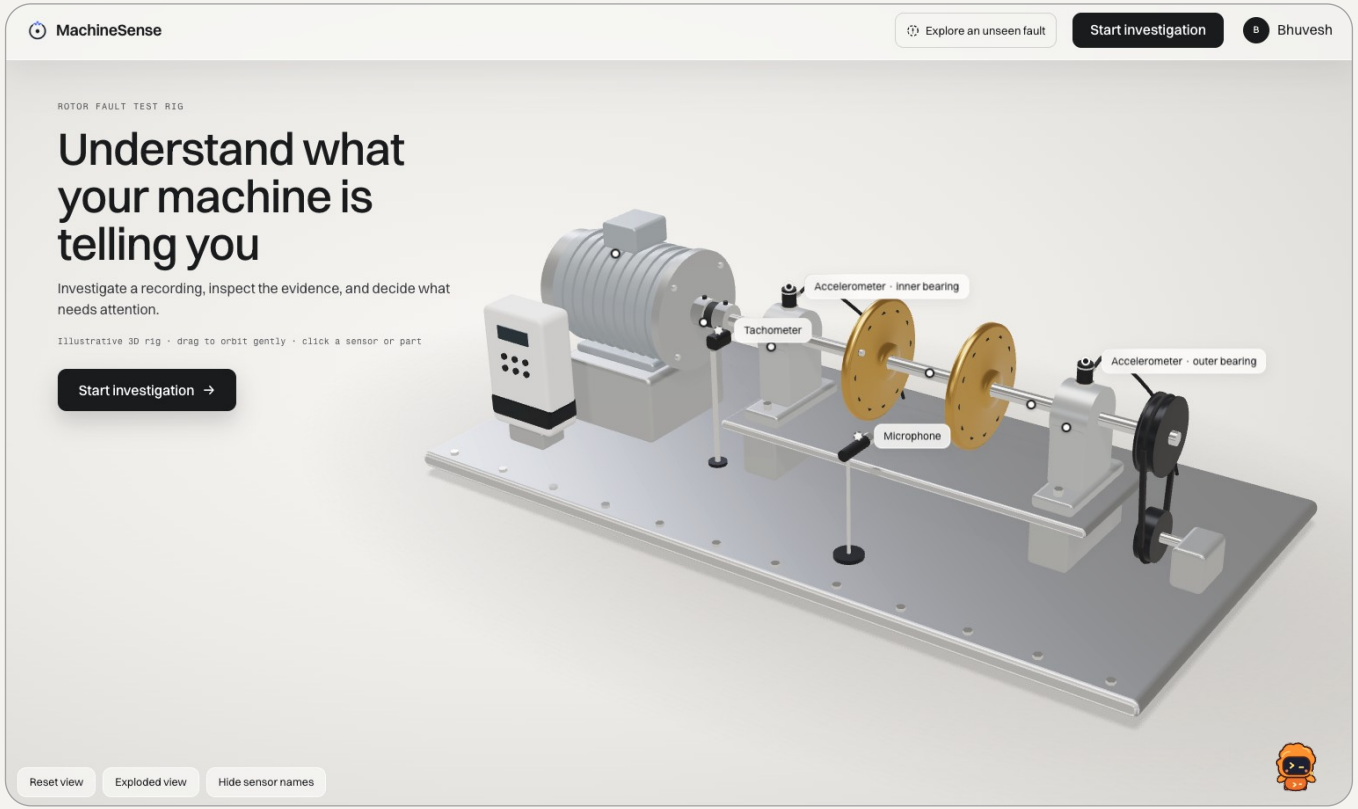
MachineSense and its objective

What is it?

MachineSense is a research prototype that uses vibration, sound, and rotation data to detect known faults and flag unfamiliar patterns. An interactive report presents the evidence, while an Investigation Assistant helps users explore the findings and ask follow-up questions.

The data used in this project.

MAFAULDA, the Machinery Fault Database from the Universidade Federal do Rio de Janeiro: 1,951 recordings of one rig run under ten deliberately induced conditions.



What MachineSense answers?

Q1

Healthy or faulty?

Detect an abnormal recording. Healthy-only checks provide a separate signal alongside the trained classifier.

Q2

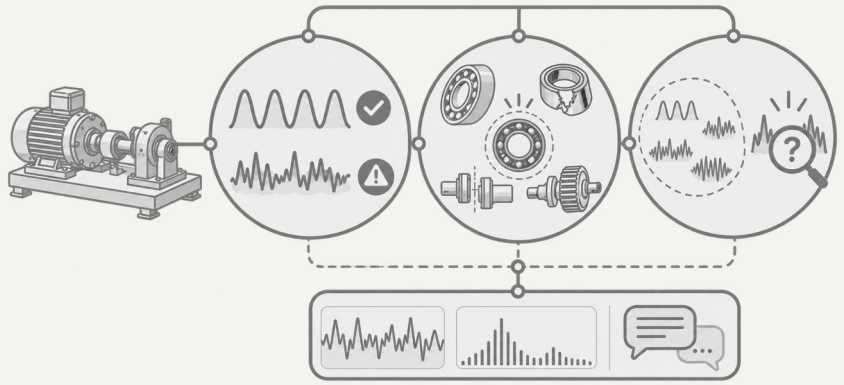
Which familiar condition?

Name a condition the classifier has learned.
(Closed-set classification)

Q3

Unfamiliar fault?

Request further inspection when a recording does not fit the learned conditions. **(Open-set recognition)**



Every objective is studied on MAFAULDA recordings from one test rig.

Problem and solution

PROBLEM

Published scores answer different questions

Evaluation leakage: Segmenting a recording before splitting can put related pieces in both training and testing. Such scores can overstate performance on new recordings.

Operating conditions: Speed and fault setting can change the signal. Random partitions alone do not show how a model behaves at unseen settings.

Unknown faults: A closed-set classifier must choose a learned class, even when a recording comes from a condition it has not seen.

Explanation gap: A mechanic needs the evidence and uncertainty behind a suggested condition.

SOLUTION

MachineSense evaluated with stricter rig tests

Recording-level, folder-grouped splits, with reserved partitions for held-out severity and held-out speed, sealed until a single declared final run.

A hybrid approach: MachineSense combines physics-based features and a classical reference model with a CNN trained on order spectrograms. A healthy-only anomaly detector and a distance-based check help flag recordings that may not match any known fault.

Closed set and open set are both reported, so declining to name a fault is a measured behaviour rather than a missing capability.

A saved evidence report that a mechanic can inspect. A bounded investigation assistant is proposed for follow-up questions.

Literature review and benchmarking

Earlier research shows that machine signals can classify familiar faults**. MachineSense asks an additional question**: when should the system stop naming a fault and request inspection?

What the field has established

Vibration and acoustic signals separate the standard rotating-machinery faults well, with both engineered features and learned representations, on this dataset and its neighbours.

Where the reported numbers break down

Segmentation before splitting can put related parts of one recording into different folds. Published scores require careful reading of each study's split and class definitions.

What is rarely asked

Most studies emphasize familiar-fault classification. MachineSense also measures the cost of declining to name an unfamiliar condition.

Benchmarking principle

Windows from the same recording stay together and results are reported per recording. Held-out operating settings test how performance changes beyond familiar examples.

On comparing numbers

Published accuracy figures should not be ranked against MachineSense unless the classes and the evaluation protocol match. Where they do not, the comparison is stated as unavailable rather than made anyway.

Where MachineSense differs

Rotating-machinery fault diagnosis on this benchmark and its neighbours.

WORK	DATA	SPLIT UNIT	OPEN SET	EXPLANATION
Ribeiro et al., SBrT 2017 similarity-based models	MAFAULDA	random sample split	no	none
Souza et al., Comp. & Ind. Eng. 2021 deep learning for rotating machinery	MAFAULDA	different protocol	no	none
Nascimento et al., arXiv 2022 T4PdM, transformer	MAFAULDA subset	different protocol	no	none
Ignjatovska et al., Sensors 2026 physics-informed feature sets	MAFAULDA	normal segments cross folds	no	feature sets only
Hendriks et al., MSSP 2022 · Matania et al., PHME 2024	CWRU and others	the subject of the work	not addressed	none
Hendrycks 2017 · Lee 2018 · Geng 2021 open-set recognition	vision benchmarks	not applicable	the subject	none
MachineSense	MAFAULDA	recording, folder-grouped	measured	per-value provenance

Published accuracy figures cannot be ranked against MachineSense without **matching classes and splits.** **My contribution is a rigorously evaluated fault-diagnosis system that keeps recordings separate, tests unseen operating settings, and flags unfamiliar faults for further inspection.**

Project scope

MachineSense is a research-demo web application that helps a mechanic interpret machine vibration and sound recordings. It studies when to report a familiar condition and when to say further inspection needed.

IN SCOPE

- Preparing and checking sensor recordings
- Detecting healthy versus faulty operation
- Naming familiar machine conditions
- Flagging unfamiliar patterns for inspection
- Showing the evidence behind each result
- A preview of a bounded investigation assistant agent
- Interactive report
- Mechanic vs Developer view (UI/UX preferences and research)

OUT OF SCOPE

- Predicting remaining useful life
- Confirming that a repair fixes the problem
- Streaming live machine data or controlling equipment
- Establishing reliability on machines other than this rig
- Replacing the mechanic's judgement
- Independent diagnosis of simultaneous faults

MachineSense supports human inspection. Further inspection needed is an uncertainty verdict, not confirmation of a new fault.

Techniques, limitations and constraints

TECHNIQUES

Supervised: a convolutional network on order spectrograms, and logistic regression on physics features

Unsupervised: an autoencoder and a speed-aware baseline fitted on healthy recordings only

Open set: Mahalanobis distance on the network's own embeddings

Hybrid: a calibrated rule combines classifier confidence, embedding distance and healthy checks

Intelligent sensing: fusion across six vibration axes, a tachometer and a microphone

LIMITATIONS

One test rig: results cannot establish reliability on other machines

Healthy recordings are scarce, so any false-alarm rate is known only to a wide interval

Open-set trade-off: Catching more unfamiliar faults also risks rejecting more known faults. We show performance across decision thresholds, as well as the chosen operating point.



CONSTRAINTS

Input format: eight-channel, five-second recordings sampled at 50 kHz, in the MAFAULDA layout

Signal quality: unreliable speed estimates or sensor readings weaken interpretation, and the application shows a warning

Deployment: the demo accepts prepared recordings and CSV uploads; it does not stream live data or control equipment

What arrives on disk

One archive, six top-level folders, 1,951 CSV files, no index and no label file.

https://www.kaggle.com/datasets/vuxuancu/mafaulda-full

1,951
CSV recordings

1 Open the folders

Folder names supply the condition label.

- **mafaulda/** — 1,951 CSVs
  - **normal/** — 18.432.csv — OPEN →
  - **imbalance/** — 30g/ → 49.9712.csv
  - **horizontal-misalignment/** — 0.5mm/ → 60.6208.csv
  - **vertical-misalignment/** — 1.78mm/ → 60.6208.csv
  - **underhang/** — outer_race/ → 0g/ → 49.9712.csv
  - **overhang/** — ball_fault/ → 6g/ → 35.6352.csv

FOLDER = LABEL
Nested folders show the setting or bearing subtype.

2 Open normal/18.432.csv

One CSV file = one five-second recording.

Raw CSV: no header row. Column names below are added for teaching.

	ROTATION	INNER BEARING · VIBRATION			OUTER BEARING · VIBRATION			SOUND
row	Tacho 1	Axial 2	Radial 3	Tangential 4	Axial 5	Radial 6	Tangential 7	Mic 8
1	−0.5703	0.026983	−0.23011	−0.016556	−0.27112	0.039268	−0.16774	0.35906
2	−0.49894	−1.9155	0.043339	0.016509	−0.28677	0.037657	−0.21446	0.034221
3	−0.59204	0.98102	0.043249	0.06109	−0.19626	0.036771	−0.13754	0.10159
⋮			more readings in the same eight columns					
250,000	…	…	…	…	…	…	…	…

ONE ROW = ONE MOMENT
All eight sensor channels were measured at that same instant.

KEY IDEA
The condition label comes from the folder path, not from a CSV column.

50,000 rows / second
× 5 seconds = 250,000 rows

Imbalance: one clean peak at shaft rate

imbalance / 35 g / 25.08 Hz measured

Left chart. The sensor voltage over time. One horizontal unit is a fraction of a second. Its shape tells you whether the machine is swaying smoothly or being struck.

Right chart. The same signal split into frequencies, but the axis is divided by the measured shaft speed. **1×** means once per turn of the shaft, **2×** twice per turn, and so on. A peak at a whole number means something happens at that rate every revolution.

Why divide by speed. A fault at 3× sits in the same place whether the shaft turns at 15 Hz or 60 Hz, so one picture serves every speed in the dataset.

Raw vibration, 80 ms of the five seconds



A smooth wave that rises and falls once per revolution. No sharp edges anywhere.

Energy against multiples of shaft speed



Almost all the energy sits at 1×. The higher orders are close to the floor.

What causes it

Extra mass on one side of the rotor pulls the shaft off centre. That pull rotates with the shaft, so it loads the bearings once per revolution.

What it looks like in numbers

High overall level, a dominant 1× peak, a low 2× to 1× ratio, and negative excess kurtosis because the signal is smoother than random noise.

Why it is the easy case

The signature is one number in one place. Both the classical model and the network separate it with almost no error on development folds.

Bearing fault: impacts across several orders

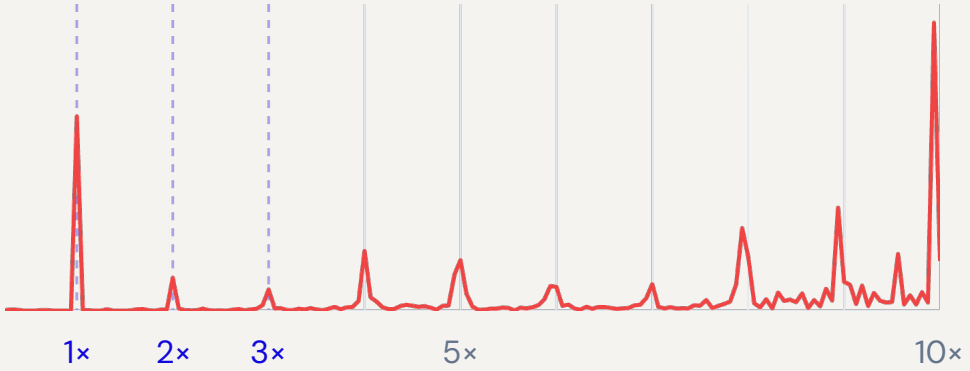
overhang / ball_fault / 0 g / 25.20 Hz
measured

Raw vibration, 80 ms of the five seconds



Sharp spikes appear on the background. They are consistent with repeated contact involving the defect.

Energy against multiples of shaft speed



Energy spread across many orders and rising toward the top, rather than concentrated anywhere.

What causes it

A pit or spall on a rolling element, a raceway or the cage. Every time it passes through the loaded zone it strikes, and the bearing rings.

Why the peak is not at a whole number

A rolling element does not complete one turn per shaft revolution, so its defect rate is a non-integer multiple of shaft speed that depends on the bearing geometry.

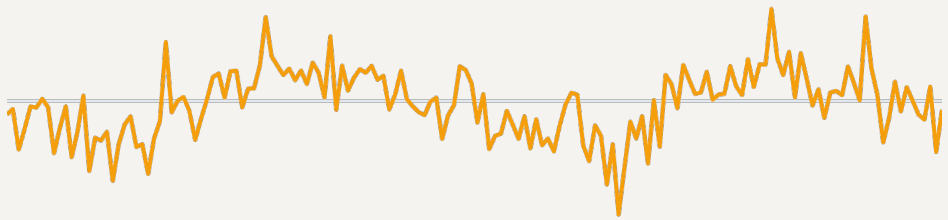
How it is actually confirmed

Not from this chart. The impacts modulate a high-frequency resonance, so the evidence comes from the envelope spectrum, where the defect rate appears as a comb of lines.

Misalignment example: energy at 2×

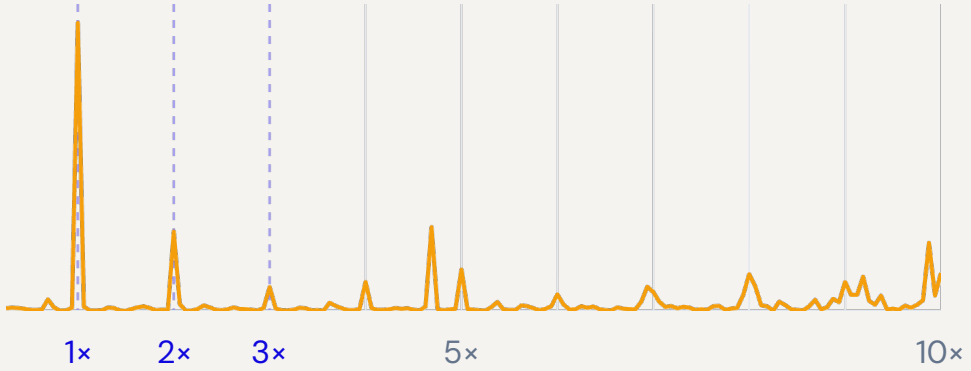
horizontal-misalignment / 2.0 mm / 25.02 Hz measured

Raw vibration, 80 ms of the five seconds



Smooth like imbalance, but the repeating unit is shorter: the pattern completes twice per revolution.

Energy against multiples of shaft speed



A visible component at 2× beside the one at 1×. That ratio is the main separator from imbalance.

What causes it

The two shafts meeting at the coupling are offset or angled. The coupling is loaded and unloaded twice per revolution rather than once.

The axial giveaway

An angular misalignment pushes along the shaft as well as across it, so the axial channels carry a component that imbalance does not produce.

Where the system still struggles

Horizontal and vertical misalignment can share this signature. Their direction-sensitive channels help distinguish them, but confusion remains possible.

Data intake and audit

1 Download & unpack

MAFAULDA is already collected.



mafaulda/

- normal/
- imbalance/
- horizontal-misalignment/
- vertical-misalignment/
- underhang/ and overhang/

1,951 CSV recordings

2 Catalogue the files

The path gives each file a label.

```
mafaulda/
└ imbalance/
  └ 30g/
    └ 49.9712.csv
```

Condition: imbalance

Setting: added mass 30 g

The filename gives nominal shaft speed in Hz.

3 Read & audit each CSV

A file is a table of raw numbers.

row	1	2	3	4	5	6	7	8
1	−.77	.11	2.43	.14	.12	.07	.31	−.20
2	−.76	−.01	1.40	.12	.11	.06	.22	.23
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

250,000 rows × 8 channels

Audit checks

- ✓ Shape and numeric values
- ✓ Sensor signal quality
- ✓ Warnings kept for review

CSV has no header row.

4 Save checked outputs

Keep the original CSV unchanged.



Numerical copy

250,000 × 8 values for fast reuse



Metadata table

Path label, speed and quality notes

Ready for split assignment →

Split, clean and window

5 Assign whole-recording splits

Decide where each CSV belongs before extracting features.

One 5-second CSV recording

Assign the recording to one set:

- TRAIN
- VALIDATION
- RESERVED TEST

All nine windows inherit this split.
Experiment-specific groups are assigned at recording level too.

6 Clean the full recording

Use settings chosen on permitted development data.



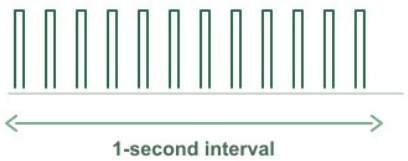
Fixed cleaning steps

1. Handle microphone bursts
2. Reduce electrical interference
3. Apply the chosen bandwidth cap

Tachometer is kept untouched.
Same procedure later for test data.

7 Estimate speed per window

Use the planned 1-second positions on the untouched tachometer.



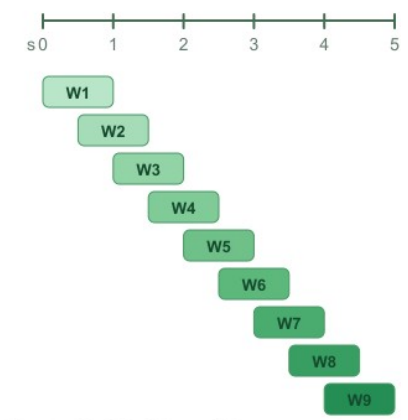
Pulse spacing → shaft speed

Window 1 → speed 1
Window 2 → speed 2
… Window 9 → speed 9

Too few clean pulses?
Use recording speed; flag fallback.

8 Cut nine overlapping windows

Each 5-second recording gives nine 1-second pieces.



Starts: 0, 0.5, 1.0, …, 4.0 s

50,000 rows × 8 channels each

Model input preparation

9

Create two feature views

Each cleaned 1-second window goes down both paths.

ONE WINDOW

PHYSICAL
142 numbers
per window

RMS · peaks · orders

IMAGE
7 maps
per window

6 vibration order spectrograms + 1 mic log-Mel

Same recording ID and window
number connect the two views.

10

Save prepared inputs

Each row or image keeps its identity, label and split.

ILLUSTRATIVE INDEX

recording	window	label	split
R042	01	imbalance	train
R042	02	imbalance	train
R043	01	normal	val

Physical features
142 values per indexed window

Spectrogram images
7 maps per indexed window

Prepared and traceable · no model fit yet

11

Fit normalisation per run

Only recordings permitted for fitting can set these values.

TRAIN **fit recordings only**

No validation or reserved test values.

LEARNED SETTINGS

centre
+ spread

per input route

Freeze the fitted settings.
A different experiment may fit different settings.

12

Apply and verify

Use the frozen settings on data the model has not fitted.

FROZEN NORMALISATION

same centre + spread

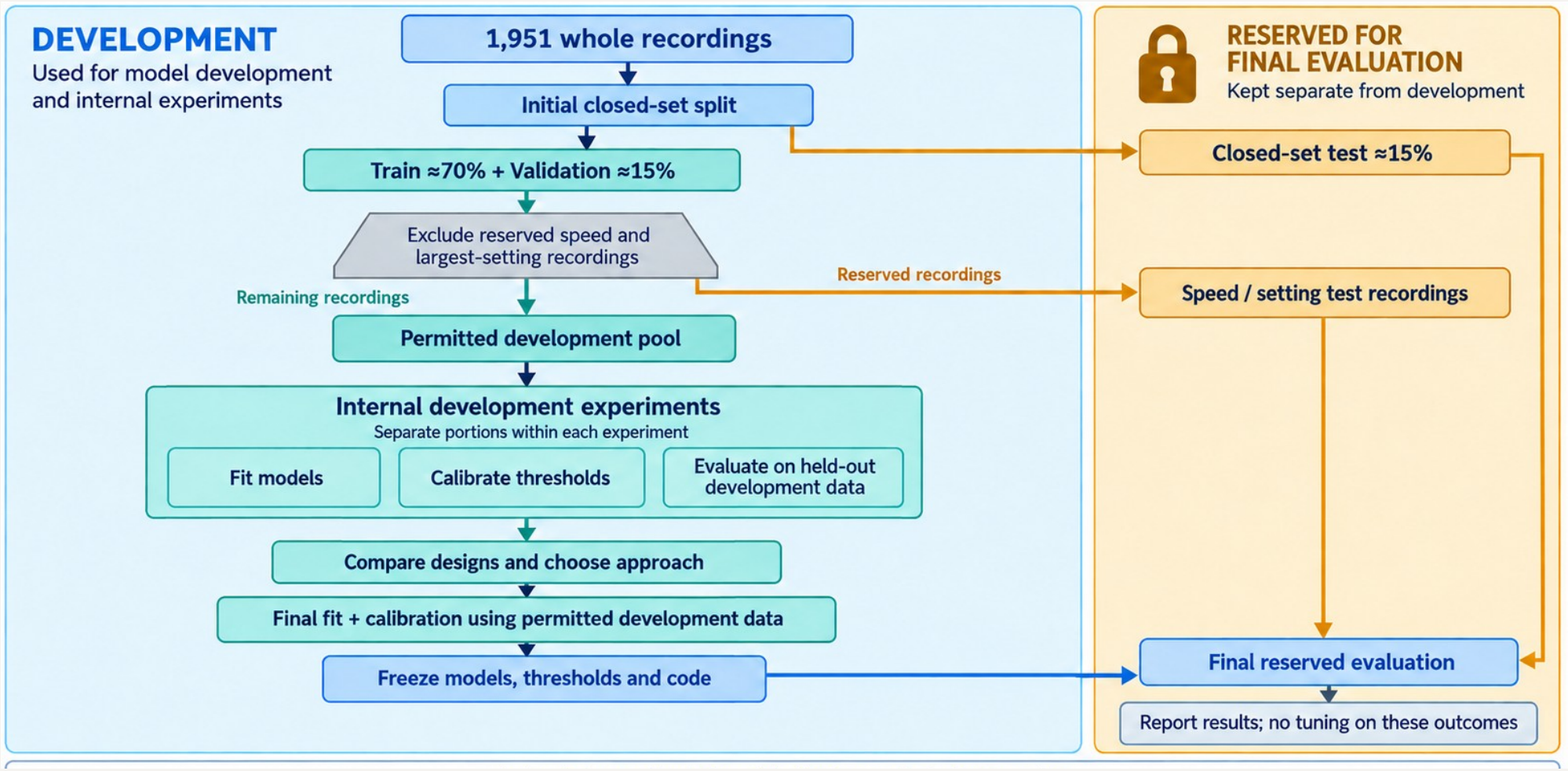
VALIDATION
tune and compare models

RESERVED TEST
use at final evaluation

Check shapes, labels, splits
and saved versions.

READY FOR MODELLING

Another view – to show how data leakage is prevented



Challenges in the data itself

Recording glitch linked to certain fault labels.

164 files carry microphone bursts of five to ten samples reaching about 160 V, against a median level of 0.13 V. They appear only in imbalance at 20 to 35 g and underhang cage at 6, 20 and 35 g, so a model could score well by detecting the artefact instead of the fault.

Tachometer glitches

228 files show double triggers or dropouts, severe in 94 of them and concentrated in underhang ball. Since every order feature depends on the shaft speed, a bad speed corrupts the whole feature vector.

Saturation and drift

Nine overhang ball recordings clip at high speed, and 44 files carry up to 2.2 V of accelerometer offset. Both look like signal unless they are checked for.

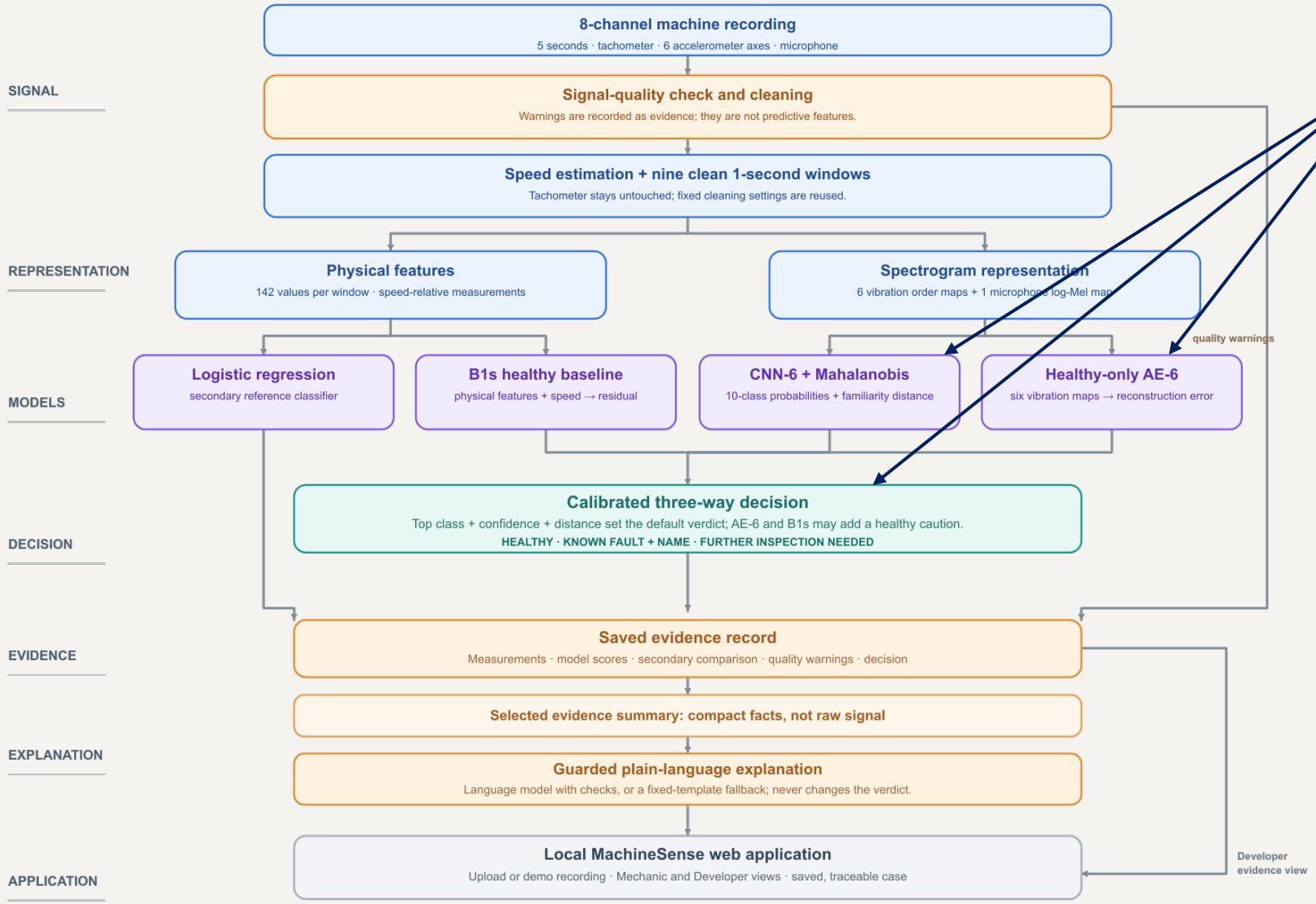
Healthy data is scarce

49 normal recordings from one session. Every false-alarm figure this project reports rests on a few dozen recordings, and no threshold choice can narrow that interval.

Added mass is not defect size

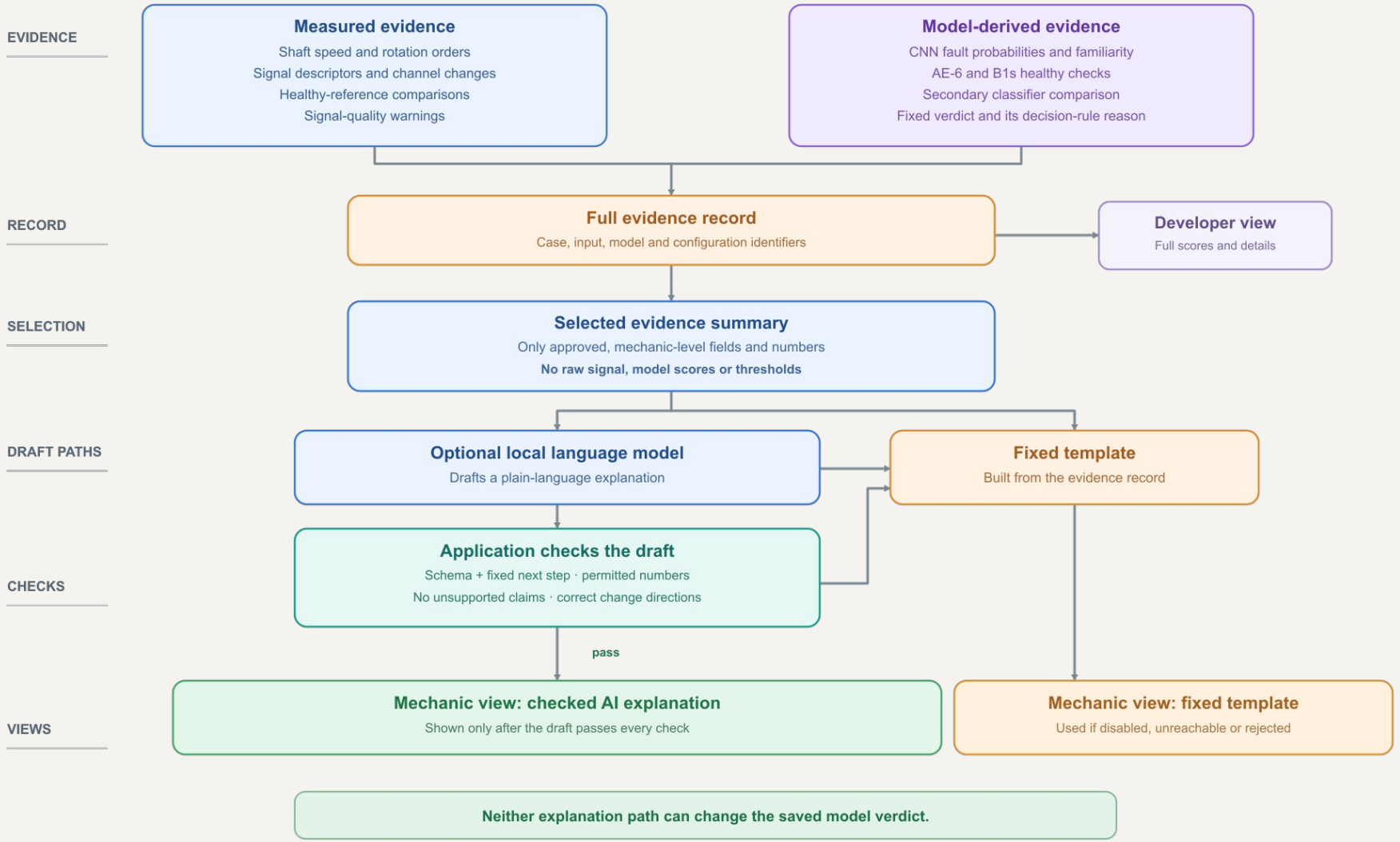
In a bearing folder, 20 g is imbalance mass sitting on top of a defect whose severity is fixed. Only in the imbalance folder does the gram figure mean severity, which is easy to misread as a severity axis across the whole dataset.

System architecture

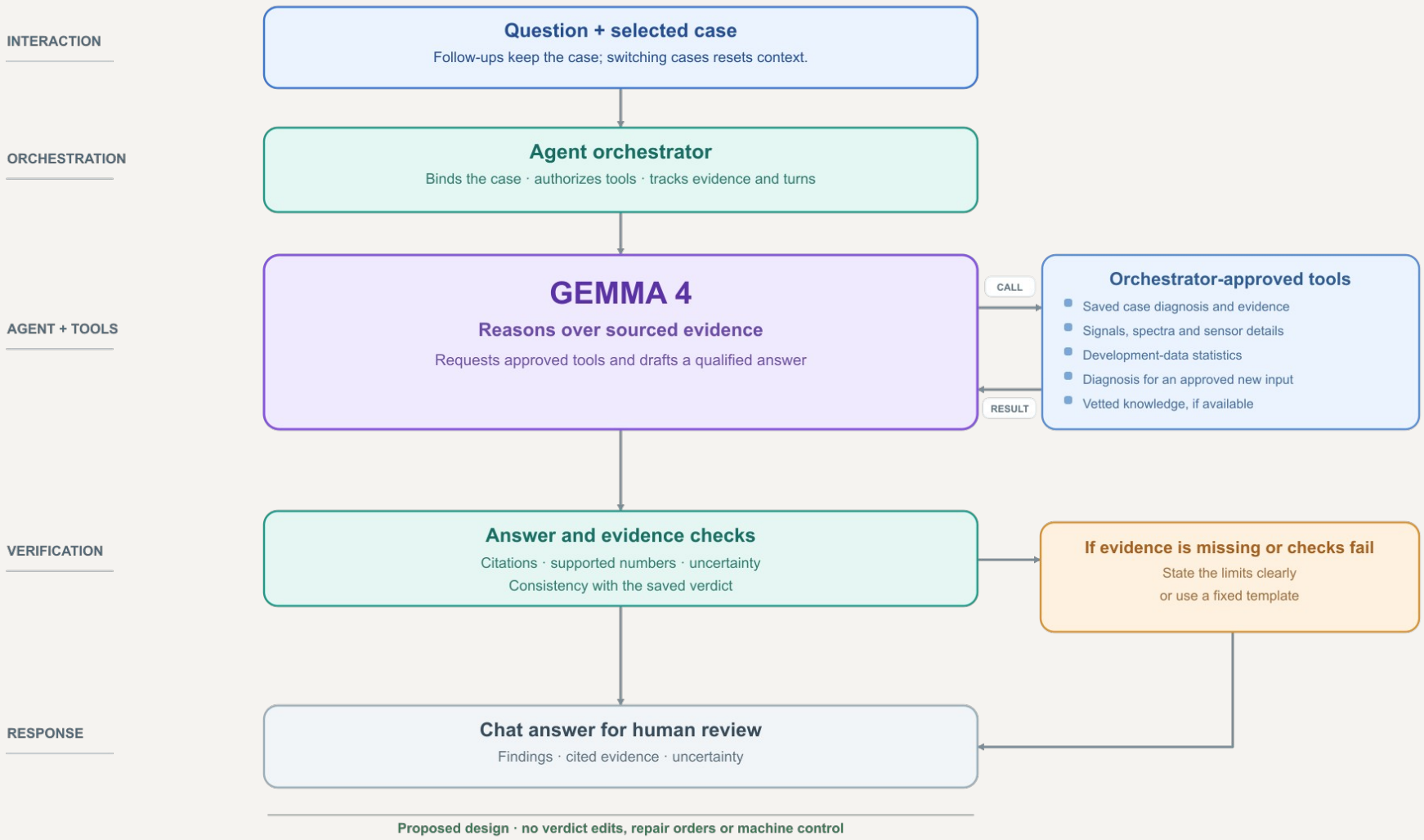


Experimentations to improve the eval metrics and design an appropriate system

Explanation architecture



Investigation assistant agent architecture



Final evaluation results

Reserved evaluation completed once. Rates below identify their test population.

Question	Evaluation metric	Results
Q1 · healthy or faulty	Fault detection rate (recall) Healthy false-alarm rate	Fault recall: **283/283 (100%)**; healthy false-alarm rate: **5/7 (71.4%)**
Q2 · familiar condition	Macro-F1, closed test Macro-F1, held-out severity Confusion matrix	Closed-test macro-F1: **0.916**; held-out-severity macro-F1: **0.747**
Q3 · unfamiliar condition	Unknown recall Known-side rejection rate	Unknown recall: **925/1,123 (82.4%)**; known-side rejection: **765/2,327 (32.9%)**

Eight attempts, one confirmed gain

Each declared in writing with its acceptance bar before it ran. Development data only.

STAGE	WHAT WAS TRIED	WHAT WAS OBSERVED	READING
D1	Learned combiner over the four decision scores	no gain over fixed thresholds	does not help
ML-5	ResNet-18 transfer learning	no gain over the 0.6 M-parameter CNN-6	does not help
R1	Speed-conditioned healthy thresholds	autoencoder AUROC 0.986 → 0.967	does not help
R3	Supervised contrastive encoder	macro-F1 −0.014; errors 8/11/4 → 21/19/14 of 804	does not help
R4	Bank of per-condition autoencoder specialists	recall tied (0.583 vs 0.582); 92 vs 19 unfamiliar called healthy	too thin to tell
E1	Healthy check first, then fault specialists only	**+10.1 points** mean recall, all 3 seeds; healthy calls unchanged	**helps, confirmed**
R6	Augmented, denoising and fault-only autoencoders	−1.1 to 0.0 points; fault-only AUROC 0.55	does not help
R7	Diagnostics on saved models, no training	mild misalignment covered by the gate; no collapse; max-logit +1.3	**diagnostic**

The bar for "helps": at least 5 points more unfamiliar faults caught at the same number of known recordings rejected, with no more unfamiliar faults called healthy, on every one of three seeds.

Investigation assistant evaluation plan

Proposed measures for the assistant after it is connected.

Agent evaluation metric	Calculation	What it checks
Grounded-Claim Precision (%)	Factual claims supported by evidence ÷ all factual claims	Whether Gemma 4 invents facts
Citation Precision (%)	Citations that support their associated claim ÷ all citations	Whether references are meaningful
Numeric Fidelity Rate (%)	Correct numbers, units and directions ÷ all numerical claims	Whether it reports measurements accurately
Verdict Consistency Rate (%)	Answers preserving the saved verdict ÷ all answers	Whether it changes “further inspection” into a claimed diagnosis
Case Isolation Rate (%)	Conversations with no cross-case evidence ÷ case-switch tests	Whether follow-ups use the right recording
Appropriate Abstention Rate (%)	Insufficient-evidence cases where it states the limit ÷ all such cases	Whether it knows when it cannot answer
Tool Policy Violation Count	Unapproved tool calls or actions	Target: zero
Fallback Success Rate (%)	Tool/model failures handled with a clear limited answer ÷ failure tests	Whether the chat remains safe and useful
P@5	Relevant items among the top five retrieved references ÷ 5	Optional: quality of the vetted-knowledge search
P95 Response Latency	Time within which 95% of answers finish	Whether the assistant is practical to use

What has been done so far?

Prepared the data: catalogued and audited 1,951 recordings, assigned recording-level splits, cleaned signals, and created windows.

Built two input types: 142 physics-based features per window and speed-relative spectrograms.

Trained the core models: a classical reference classifier, a CNN for known faults, and a healthy-only autoencoder / Specialist AEs.

Designed the decision system: combined classification, healthy-pattern checks, and Mahalanobis distance to allow an “unfamiliar-inspect further” outcome.

Ran development experiments: known-fault classification, healthy-versus-faulty detection, and leave-one-condition-out unfamiliar-fault evaluation.

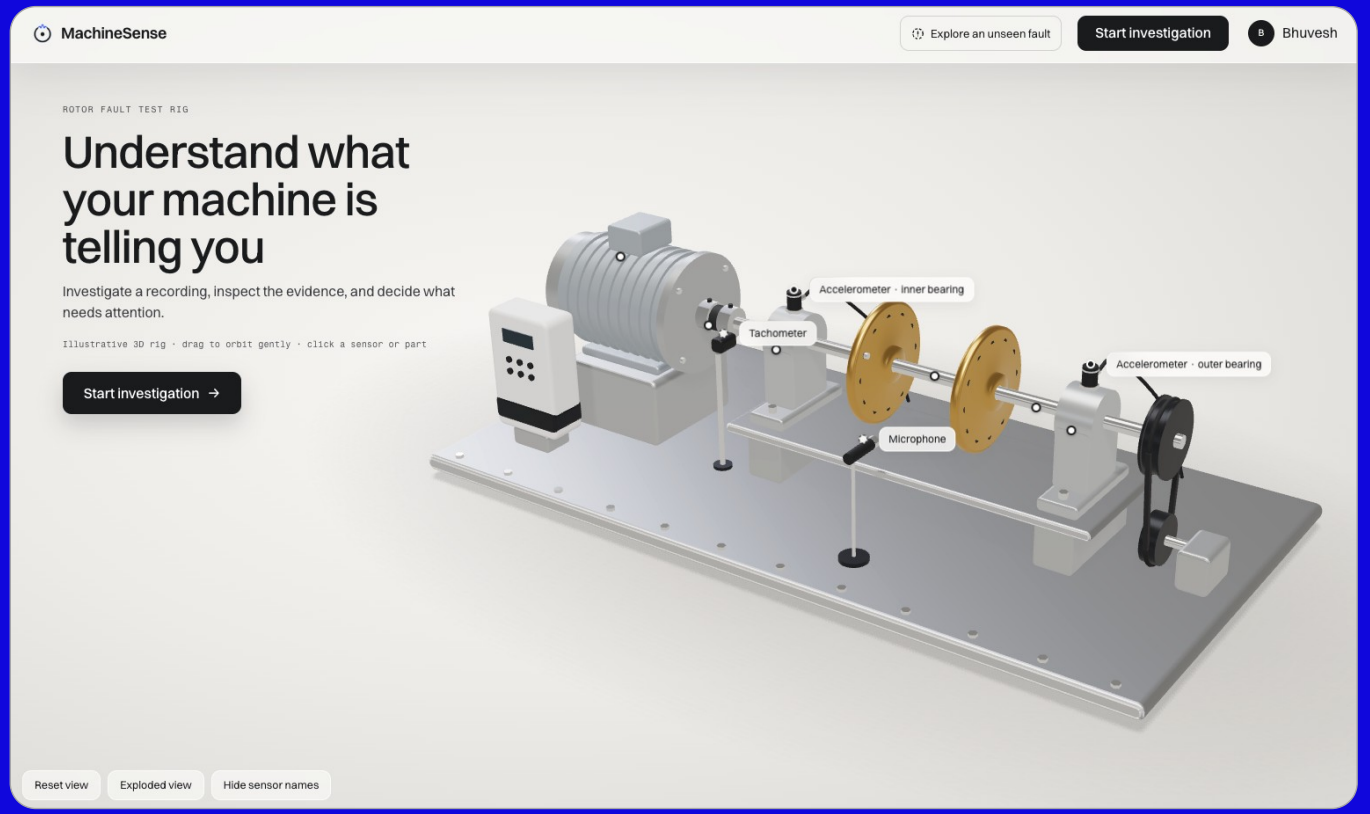
Investigated alternatives: specialist autoencoders, data augmentation, denoising, a fault-only autoencoder, alternative confidence gates, and the gated **E1** specialist approach.

Built the demonstration app: an interactive machine view and saved evidence reports. The Investigation Assistant remains proposed, not connected.

Short Demo

github.com/bhuveshsharma09/MachineSense

Bhuvesh Kumar · NUS-ISS



Say what the evidence supports, and no more

Three questions from one reading, a verdict that can decline to name a fault, and every number traceable to the stage that produced it.

github.com/bhuveshsharma09/MachineSense

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